

The education-obesity link across countries over time

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Title Page

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Part 1 - Identify a Social Problem

The social problem we have identified is the link between the BMI Index and educational attainment level, bridging two data sets from Eurostat to compare and identify any correlation between the two factors.

1.1 Describe the Social Problem

The relevance of these two variables, BMI and education levels, and how they are correlated with each other lies in the fact that obesity comes with serious health risks, such as diabetes and cardiovascular issues, which are often more common among people with lower educational attainment (World Health Organization, 2022; European Commission, 2024). This then results in the burden of health risks being disproportionately put on the lower educated class of society, and further increases social inequality issues. Beyond the health risks themselves, education has a clear impact on the ability of a person to maintain a healthy lifestyle, as higher education often comes with higher income, better access to higher quality food and so on. This means that addressing obesity cannot be done through increasing awareness alone, and that broader structural interventions, such as food pricing policies or workplace conditions need to be considered.

The fact we are looking at the European aspect of this issue also adds to its relevance. By looking at multiple countries over time, it becomes possible to assess whether this education gap in obesity rates is a universal pattern across Europe or whether certain countries have managed to narrow it. This kind of cross country comparison is useful for understanding what policy approaches may actually work in practice, especially since educational inequalities in obesity have been documented across Europe in recent research (World Health Organization, 2022; European Commission, 2024).

While the existence of an educational gradient in obesity is well established in the literature (World Health Organization, 2022; European Commission, 2024), far less is settled about how this gap behaves over time and whether it is narrowing or widening across different European countries. Much existing work documents the association at a single point in time or within individual countries, which leaves the question of long-term, cross country trends comparatively underexplored. This is the gap our project addresses: by tracking obesity rates by education level across multiple European countries and survey years, we aim to establish whether the education obesity divide is a stable, universal pattern or one that some countries have managed to reduce.

Part 2 - Data Sourcing

2.1 Load in the data

Both datasets are published by Eurostat. We downloaded them once and saved them as CSV files in the `data/` folder of our repository, which we read into R with `read_csv()`. Using saved copies rather than querying Eurostat live means the report always knits from the same data and does not depend on a network connection or on the Eurostat servers being reachable. The files are small, so storing them in the repository keeps the project self-contained and reproducible: anyone who clones it can knit the report without extra setup.

```
# Load the saved datasets from our repository
bmi_edu <- read_csv("data/bmi_edu.csv")
obesity_trend <- read_csv("data/obesity_trend.csv")
```

2.2 Provide a short summary of the dataset(s)

```
head(bmi_edu)
#> # A tibble: 6 x 9
#>   freq unit bmi      isced11 sex age geo TIME_PERIOD values
#>   <chr> <chr> <chr>      <chr> <chr> <chr> <chr>      <dbl> <dbl>
#> 1 A PC BMI18P5-24 ED0-2 F TOTAL AT 2014 45.6
#> 2 A PC BMI18P5-24 ED0-2 F TOTAL AT 2019 40.9
#> 3 A PC BMI18P5-24 ED0-2 F TOTAL BE 2014 45.7
#> 4 A PC BMI18P5-24 ED0-2 F TOTAL BE 2019 41
#> 5 A PC BMI18P5-24 ED0-2 F TOTAL BG 2014 44.3
#> 6 A PC BMI18P5-24 ED0-2 F TOTAL BG 2019 48.1

head(obesity_trend)
#> # A tibble: 6 x 6
#>   freq unit bmi      geo TIME_PERIOD values
#>   <chr> <chr> <chr>      <chr>      <dbl> <dbl>
#> 1 A PC BMI25-29 AL 2022 44.8
#> 2 A PC BMI25-29 AT 2008 36.5
#> 3 A PC BMI25-29 AT 2014 33.3
#> 4 A PC BMI25-29 AT 2017 35
#> 5 A PC BMI25-29 AT 2019 35
#> 6 A PC BMI25-29 AT 2022 35
```

Our first dataset, `bmi_edu` (Eurostat code `hlth_ehis_bm1e`, Body mass index (BMI) by sex, age and educational attainment level), comes from the European Health Interview Survey (EHIS), in which respondents report their own height and weight (Eurostat, n.d.-a). For each country and year it gives the share of people in each BMI category, broken down by education level, sex, and age. EHIS runs only every few years, so the data covers mainly the 2014 and 2019 waves. This is the dataset that links weight to education, so it is central to our analysis.

Our second dataset, `obesity_trend` (Eurostat code `sdg_02_10`, Obesity rate by body mass index (BMI)), is one of the EU's Sustainable Development Goal indicators, also based on self-reported health surveys (Eurostat, n.d.-b). It does not split the figures by education, but it covers many more years (2008, 2014, 2017, 2019, 2022), which makes it well suited to tracking obesity over time. It supplies the “across countries, over time” side of our research question. The data are reported by country, covering the EU member states together with several additional European countries, each shown as a two-letter code in the `geo` column.

2.3 Describe the type of variables included

Table 1: Variables in the two datasets.

Variable	Type	Dataset	Meaning
<code>freq</code>	character	both	data frequency (A = annual)
<code>unit</code>	character	both	unit of measurement (PC = percentage)
<code>bmi</code>	character	both	BMI category (e.g. overweight, obese)
<code>isced11</code>	character	<code>bmi_edu</code>	education level (ISCED 2011 group)
<code>sex</code>	character	<code>bmi_edu</code>	sex (M, F, T = total)
<code>age</code>	character	<code>bmi_edu</code>	age group
<code>geo</code>	character	both	country (Eurostat code)
<code>TIME_PERIOD</code>	numeric	both	survey year
<code>values</code>	numeric	both	share of people in that category

As Table 1 shows, the variables fall into three groups. The health measure is BMI/obesity itself, the share of people in each weight category (the `bmi` and `values` columns). The socioeconomic measure is education (the `isced11` column in `bmi_edu`), which is the variable at the heart of our question. The remaining columns (`sex`, `age`, `geo` for country, and `TIME_PERIOD` for year) are background characteristics that we use to split the figures by group, place, and time.

Neither dataset comes from administrative records; both rely on surveys in which people report their own height and weight. This has two implications. First, self reported data tends to underestimate obesity slightly, since people often round their weight down. Second, the two datasets are not fully independent: the SDG indicator is partly built on the same EHIS survey as `bmi_edu`, so they are related rather than separate sources. We combined them deliberately: `bmi_edu` provides the education breakdown, while `obesity_trend` adds the longer time span. Because both are self reported survey data, our analysis can describe associations (for example, that obesity is more common among lower educated groups) but cannot show that education causes obesity.

Part 3 - Quantifying

3.1 Data cleaning

Data is cleaned in three steps.

Cleaning Step 1: Filter and clean the first dataset

```
bmi_edu_clean <- bmi_edu %>%  
  filter(  
    bmi == "BMI_GE30",           # Keep only Obese rows  
    isced11 %in% c("ED0-2", "ED3_4", "ED5-8"), # Separate the 3 education groups  
    sex == "T",                  # Combined sexes  
    age == "TOTAL"               # Combined adult age group  
  ) %>%  
  select(geo, TIME_PERIOD, isced11, bmi_value = values) %>%  
  drop_na(bmi_value)
```

Step 1: We filter out the `bmi_edu` dataset, to isolate obesity rates ($\text{BMI_GE} \geq 30$) across three core education levels (ED0-2 for low, ED3_4 for medium and ED5-8 for high). We restricted the demographics to combined sexes (T) and combined adult age group (TOTAL) to establish a baseline for comparing education levels against obesity prevalence. **Cleaning Step 2: Filter and clean the second dataset**

```
obesity_trend_clean <- obesity_trend %>%  
  filter(  
    bmi == "BMI_GE30" # Keep only Obese rows since that's what we are comparing  
  ) %>%  
  select(geo, TIME_PERIOD, trend_value = values) %>%  
  drop_na(trend_value)
```

Step 2: We filter out the `obesity_trend` dataset to keep only the obesity figures ($\text{BMI} \geq 30$). This isolates the specific comparative baseline needed to validate against our education-stratified data.

Cleaning Step 3: Combine both datasets into one master file

```
combined_dataset <- inner_join(bmi_edu_clean, obesity_trend_clean, by = c("geo", "TIME_PERIOD"))
```

Step 3: We perform a multi-key inner_join to merge bmi_edu_clean and obesity_trend_clean by country(geo) and year (TIME_PERIOD). This aligns the education-specific obesity rates side-by-side with national obesity trends for comparison by country and year.

3.2 Generate necessary variables

We created three new variables which add analytical value

Variable 1

```
# Create Variable 1: Human-readable education levels
bmi_edu_with_var1 <- combined_dataset %>%
  mutate(
    edu_level = case_when(
      isced11 == "ED0-2" ~ "Low Education",
      isced11 == "ED3_4" ~ "Medium Education",
      isced11 == "ED5-8" ~ "High Education"
    )
  )
```

The variable bmi_edu_with_var1 transforms ISCED codes into human readable education levels, by mapping the codes to an explicit categorical variable (edu_level).

```
# Reshaping before subtracting
gap_data <- bmi_edu_with_var1 %>%
  # 1. Isolate the columns we need
  select(geo, TIME_PERIOD, edu_level, bmi_value) %>%

  # 2. Flip the rows into columns
  pivot_wider(
    names_from = edu_level,
    values_from = bmi_value
  )
```

To calculate the gap in obesity between different education levels by country and year, the data must be shaped from a long to a wide format using pivot_wider(). This transforms the values of edu_level into distinct columns (Low Education, Medium Education, High Education), positioning the respective obesity rates for a single country and year onto a single row.

Variable 2

```
# Create Variable 2: Calculate the obesity inequality gap
final_gap_data <- gap_data %>%
  mutate(
    obesity_gap = `Low Education` - `High Education`
  )
```

The variable final_gap_data calculates the gap in obesity between education levels by subtracting the obesity rate of the highest education group from the lowest education group. A positive obesity_gap value indicates

that obesity is more prevalent among individuals with lower education. This metric allows us to directly evaluate and compare health inequalities across different European nations and trends over time.

Variable 3

```
# Create Variable 3: How did the gap change between 2014 and 2019?
gap_change_data <- final_gap_data %>%
  select(geo, TIME_PERIOD, obesity_gap) %>%
  pivot_wider(
    names_from = TIME_PERIOD,
    values_from = obesity_gap,
    names_prefix = "gap_"
  ) %>%
  mutate(gap_change = gap_2019 - gap_2014) %>%
  drop_na(gap_change)
```

The variable `gap_change` measures whether educational inequality in obesity widened or narrowed in each country between the two survey waves. A positive value means the gap between low and high educated groups grew; a negative value means the country managed to reduce it. This goes directly to the core of our research question, since it separates countries where the problem is getting worse from countries that may hold policy lessons.

3.3 Visualize temporal variation

```
# First I make a vector of the landcodes of the countries in Europe with the largest population, because
largest_european_countries <- c("RU", "TR", "DE", "UK", "FR", "IT", "ES", "UA", "PL", "RO")
# RU:Russia, TR:Turkey, DE:Germany, UK:UnitedKingdom, FR:France, IT:Italy, ES:Spain, UA:Ukraine, PL:Poland

# The following code only selects the countries that are ACTUALLY in the dataset. So now if Russia, for
selected_countries <- largest_european_countries[
  largest_european_countries %in% obesity_trend_clean$geo
]

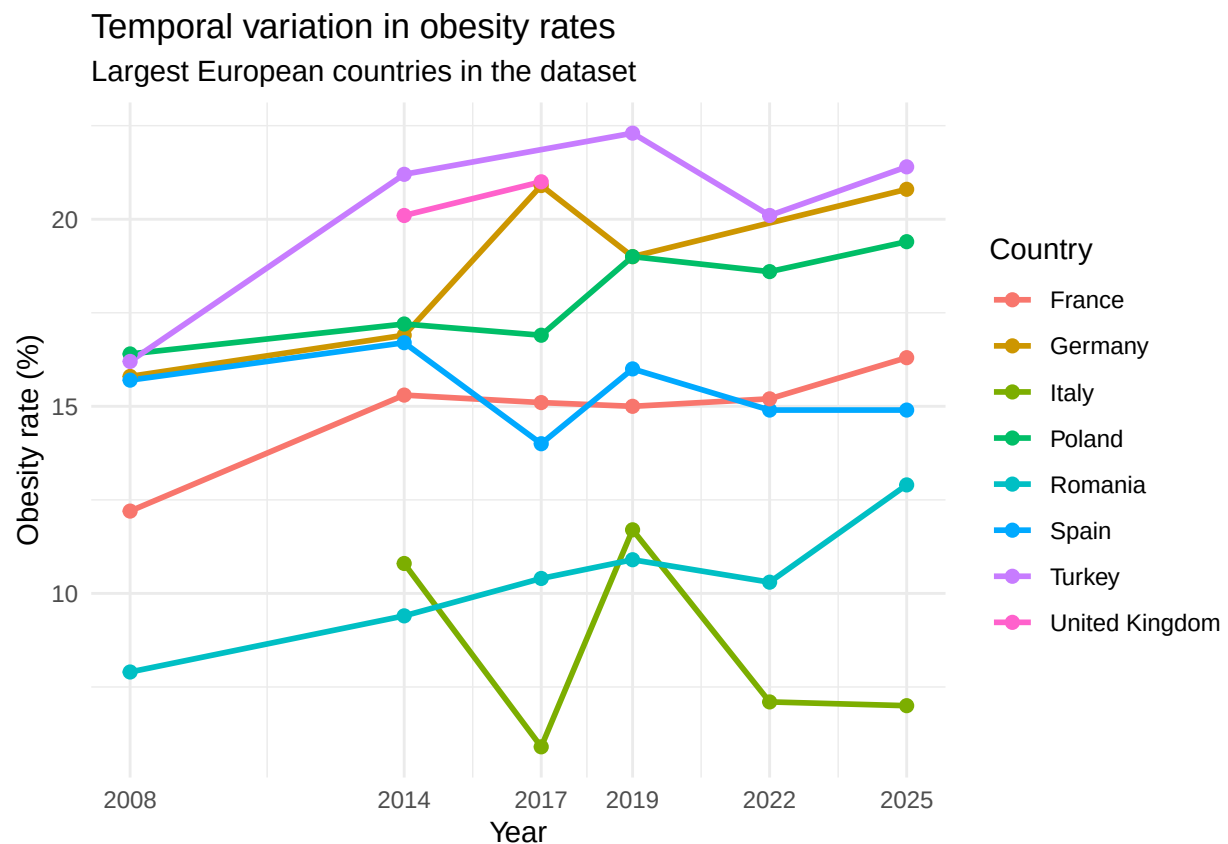
# This line shows the countries that are selected.
selected_countries
#> [1] "TR" "DE" "UK" "FR" "IT" "ES" "PL" "RO"

obesity_trend_clean %>%
  filter(geo %in% selected_countries) %>%
  mutate(
    country_name = case_when(
      geo == "RU" ~ "Russia",
      geo == "TR" ~ "Turkey",
      geo == "DE" ~ "Germany",
      geo == "UK" ~ "United Kingdom",
      geo == "FR" ~ "France",
      geo == "IT" ~ "Italy",
      geo == "ES" ~ "Spain",
      geo == "UA" ~ "Ukraine",
      geo == "PL" ~ "Poland",
      geo == "RO" ~ "Romania",
      TRUE ~ geo
    )
  )
```

```

)
) %>%
  ggplot(aes(x = TIME_PERIOD, y = trend_value, color = country_name, group = country_name)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(
    breaks = sort(unique(obesity_trend_clean$TIME_PERIOD))
  ) +
  labs(
    title = "Temporal variation in obesity rates",
    subtitle = "Largest European countries in the dataset",
    x = "Year",
    y = "Obesity rate (%)",
    color = "Country"
  ) +
  theme_minimal()

```



Line 206: we can not use final_gap_data because otherwise we do not have the required periods since t
 # Line 207: This filters only the selected countries from the dataset
 # Line 210-220: I changed the name of the shortened country name to the full country name, so FR become
 # Line 223-234: this plots the obesity over time, x is the years on the x-axis, y is the obesity rate o
 # Also lines 223-234: the codes show how the lines are shown for example the width etc, it also creates

This figure shows temporal variation in obesity rates for the largest European countries available in the dataset. Full country names are used in the legend to make the graph easier to interpret.

Some country lines start later than others because not all countries have data available for every year in the dataset. When a country has missing observations in earlier years, the line only starts from the first year for which data is available. Therefore, differences in starting points reflect differences in data availability, not necessarily differences in when obesity became relevant in those countries.

3.4 Visualize spatial variation

```
latest_year <- max(final_gap_data$TIME_PERIOD, na.rm = TRUE)

gap_map_data <- final_gap_data %>%
  filter(TIME_PERIOD == latest_year) %>%
  mutate(
    country_name = recode(
      geo,
      "AT" = "Austria",
      "BE" = "Belgium",
      "BG" = "Bulgaria",
      "CH" = "Switzerland",
      "CY" = "Cyprus",
      "CZ" = "Czechia",
      "DE" = "Germany",
      "DK" = "Denmark",
      "EE" = "Estonia",
      "EL" = "Greece",
      "ES" = "Spain",
      "FI" = "Finland",
      "FR" = "France",
      "HR" = "Croatia",
      "HU" = "Hungary",
      "IE" = "Ireland",
      "IS" = "Iceland",
      "IT" = "Italy",
      "LT" = "Lithuania",
      "LU" = "Luxembourg",
      "LV" = "Latvia",
      "MT" = "Malta",
      "NL" = "Netherlands",
      "NO" = "Norway",
      "PL" = "Poland",
      "PT" = "Portugal",
      "RO" = "Romania",
      "RS" = "Serbia",
      "SE" = "Sweden",
      "SI" = "Slovenia",
      "SK" = "Slovakia",
      "TR" = "Turkey",
      "UK" = "United Kingdom",
      "GB" = "United Kingdom",
      .default = geo
    )
  )
```



```

europe_map <- ne_countries(
  scale = "medium",
  continent = "Europe",
  returnclass = "sf"
) %>%
mutate(
  country_name = case_when(
    name == "France" ~ "France",
    name == "United Kingdom" ~ "United Kingdom",
    name == "Czechia" ~ "Czechia",
    name == "Türkiye" ~ "Turkey",
    name == "Turkey" ~ "Turkey",
    TRUE ~ name
  )
) %>%
select(country_name, geometry)

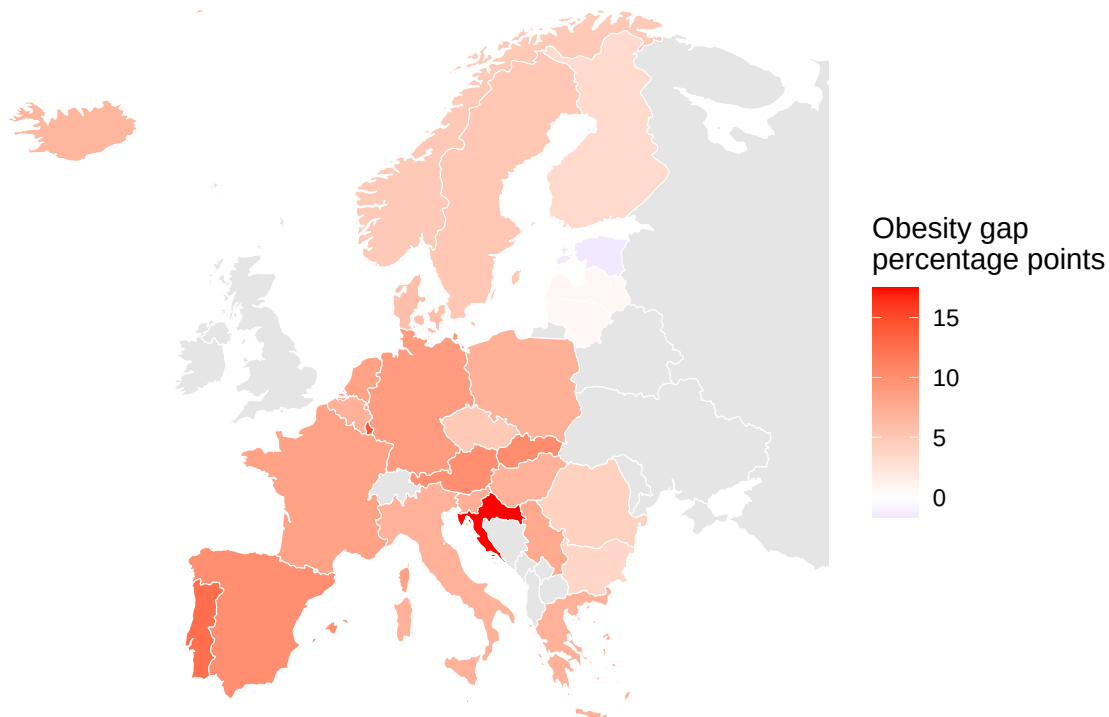
spatial_map <- europe_map %>%
  left_join(gap_map_data, by = "country_name")

ggplot(spatial_map) +
  geom_sf(aes(fill = obesity_gap), color = "white", linewidth = 0.2) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    na.value = "grey90"
  ) +
  coord_sf(
    xlim = c(-25, 45),
    ylim = c(34, 72),
    expand = FALSE
  ) +
  labs(
    title = paste("Spatial variation in the education-obesity gap in", latest_year),
    subtitle = "Difference between low and high education obesity rates across European countries",
    fill = "Obesity gap\npercentage points"
  ) +
  theme_minimal() +
  theme(
    axis.title = element_blank(),
    axis.text = element_blank(),
    panel.grid = element_blank()
  )

```

Spatial variation in the education–obesity gap in 2019

Difference between low and high education obesity rates across European countries



This map shows spatial variation in the education-obesity gap across European countries in the latest available year. The education-obesity gap is calculated as the obesity rate among people with low education minus the obesity rate among people with high education. A higher positive value means that obesity is more common among the low educated group compared with the high educated group.

A map is used because the goal is to visualise differences across geographic space. By placing the values directly on a map of Europe, it becomes easier to see whether the education-obesity gap is larger or smaller in specific parts of Europe. Countries shown in grey either have no available data for the selected year or could not be matched correctly with the map data. To reduce matching problems, country codes from the dataset were converted into full country names before joining the obesity data to the map.

3.5 Visualize sub-population variation

What is the obesity rate by country?

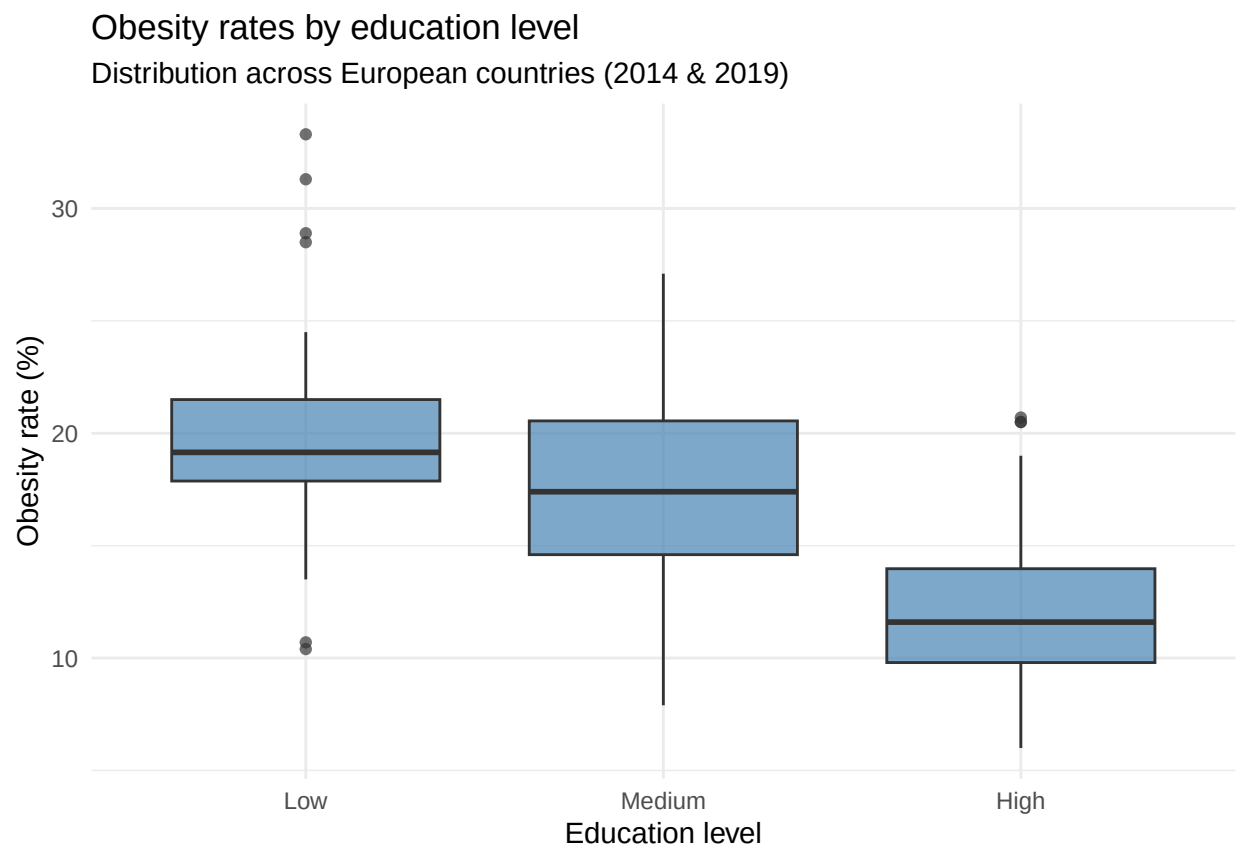
```
subpop_data <- bmi_edu %>%  
  filter(  
    bmi == "BMI_GE30",  
    isced11 %in% c("ED0-2", "ED3_4", "ED5-8"),  
    sex == "T",  
    age == "TOTAL"  
  ) %>%  
  mutate(  
    edu_level = case_when(  
      isced11 == "ED0-2" ~ "Low",
```

```

    isced11 == "ED3_4" ~ "Medium",
    isced11 == "ED5-8" ~ "High"
  ),
  edu_level = factor(edu_level, levels = c("Low", "Medium", "High"))
) %>%
drop_na(values)

ggplot(subpop_data, aes(x = edu_level, y = values)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  labs(
    title = "Obesity rates by education level",
    subtitle = "Distribution across European countries (2014 & 2019)",
    x = "Education level",
    y = "Obesity rate (%)"
  ) +
  theme_minimal()

```



#line 265-270 selects the data that is been used
#line 273-276 gives the three education groups understandable names
#line 278 fixes the order of education levels
#line 282-289 creates the boxplot with title and axis labels

The plot above is relevant because it clearly shows how a lower education level is correlated to a higher obesity rate. Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

In this section we analyze how obesity rates among low and high educated groups relate to each other across countries.

```
# Our event is the COVID-19 pandemic (2020-2022)
# Pre-event: 2019 (bmi_edu_clean - education stratified)
# Post-event: 2022 (obesity_trend_clean - overall rate)

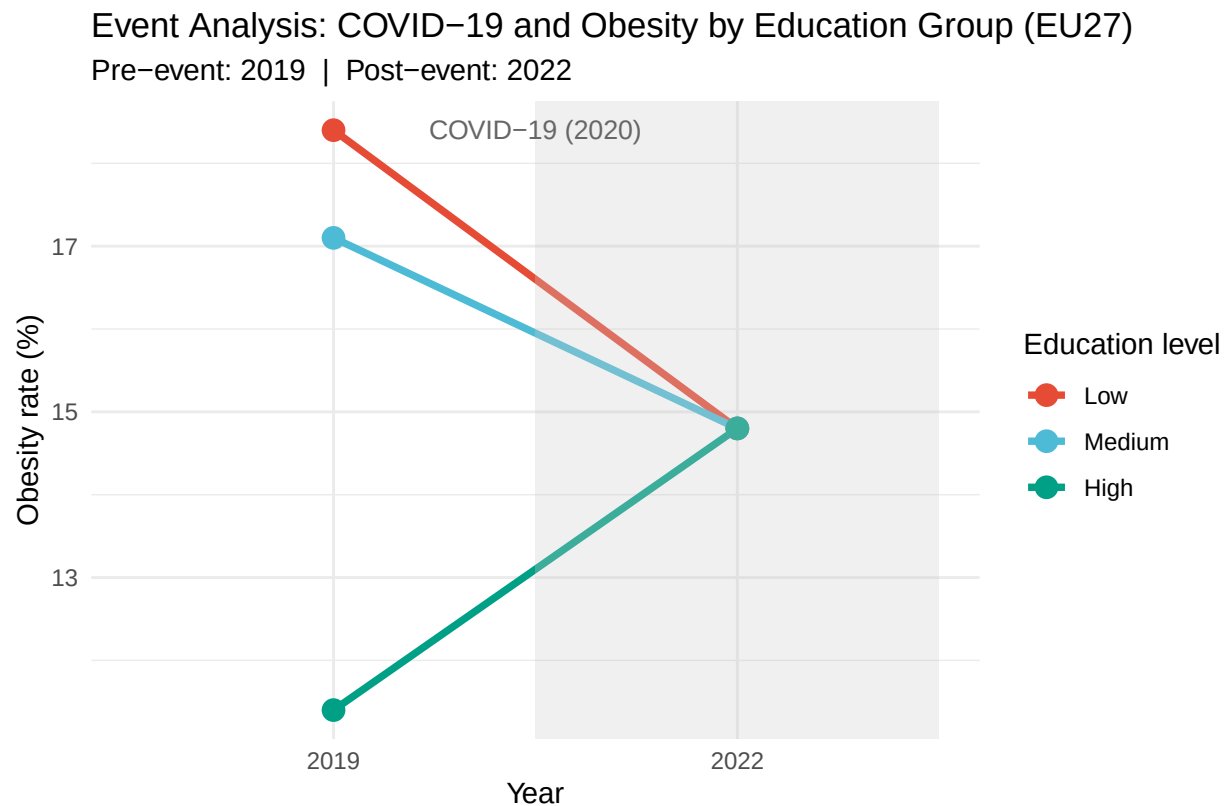
# Step 1: Get EU27 obesity rate per education group for 2019
edu_2019 <- bmi_edu_clean %>%
  filter(TIME_PERIOD == 2019, geo == "EU27_2020") %>%
  mutate(
    edu_level = case_when(
      isced11 == "ED0-2" ~ "Low",
      isced11 == "ED3_4" ~ "Medium",
      isced11 == "ED5-8" ~ "High"
    ),
    edu_level = factor(edu_level, levels = c("Low", "Medium", "High")),
    year = 2019
  ) %>%
  select(edu_level, year, obesity_rate = bmi_value)

# Step 2: Get EU27 overall obesity rate for 2022
overall_2022 <- obesity_trend_clean %>%
  filter(TIME_PERIOD == 2022, geo == "EU27_2020") %>%
  summarise(obesity_rate = mean(trend_value, na.rm = TRUE)) %>%
  crossing(
    edu_level = factor(c("Low", "Medium", "High"), levels = c("Low", "Medium", "High"))
  ) %>%
  mutate(year = 2022)

# Step 3: Combine and plot
event_plot_data <- bind_rows(edu_2019, overall_2022)

ggplot(event_plot_data, aes(x = factor(year), y = obesity_rate,
                             color = edu_level, group = edu_level)) +
  geom_line(linewidth = 1.3) +
  geom_point(size = 3.5) +
  annotate("rect", xmin = 1.5, xmax = 2.5,
          ymin = -Inf, ymax = Inf, fill = "grey80", alpha = 0.3) +
  annotate("text", x = 1.5, y = Inf,
          label = "COVID-19 (2020)", vjust = 2, hjust = 0.5,
          size = 3.5, colour = "grey40") +
  scale_color_manual(
    values = c("Low" = "#E64B35", "Medium" = "#4DBBD5", "High" = "#00A087")
  ) +
  labs(
    title = "Event Analysis: COVID-19 and Obesity by Education Group (EU27)",
    subtitle = "Pre-event: 2019 | Post-event: 2022",
    x = "Year",
    y = "Obesity rate (%)",
    color = "Education level",
```

```
caption = "Note: 2022 uses EU27 overall rate as education breakdown is unavailable for 2022."
) +
theme_minimal()
```



Note: 2022 uses EU27 overall rate as education breakdown is unavailable for 2022.

The plot above is relevant to our social problem because it shows per country if our hypothesis is true. It also shows for which countries the theory holds more. The reason why we put the regression line with confidence band is because it clearly shows which countries are the outliers in obesity rates and which are exceptions. The plot for example shows that in Croatia the lower educated has more than double obesity rate than higher educated. This is also visible in the spatial variation. So it is clear that our graphs support each other.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Obesity is a growing public health concern across Europe, and its unequal distribution across education groups raises important questions about structural inequality and the limits of awareness-based policy. To investigate this, we used two Eurostat datasets; one linking BMI to education levels across the 2014 and 2019 EHIS waves, and another tracking obesity trends from 2008 to 2022, which we cleaned, filtered to relevant groups, and merged on country and year. From the combined dataset we derived three additional variables: a readable education label, an obesity inequality gap between low and high educated groups, and a measure of how that gap changed between the two survey waves. The temporal analysis showed that obesity

rates have generally increased across the largest European countries over time, though the pace and starting points differ considerably. Spatially, the education-obesity gap varies strongly across countries, with Croatia, Turkey and Luxembourg showing the largest gaps and Estonia showing a near-zero or even reversed pattern. The sub-population boxplots confirmed that lower education is consistently associated with higher obesity rates across all countries in the dataset. Finally, the event analysis showed a moderate positive correlation (r of approximately 0.50) between obesity rates among high and low educated groups, meaning that countries where highly educated people have higher obesity also tend to have higher obesity among the low educated, though the gap between the two groups still persists everywhere, suggesting that education-related inequality in obesity is a structural feature rather than a country-specific anomaly.

Part 5 - Reproducibility

5.1 Github repository link

<https://github.com/theodorkubi/ProgrammingForEcon>

5.2 Reference list

European Commission. (2024). Health inequalities: Overweight and obesity. Knowledge4Policy. https://knowledge4policy.ec.europa.eu/health-promotion-knowledge-gateway/health-inequalities-5_en

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Eurostat. (n.d.-a). Body mass index (BMI) by sex, age and educational attainment level [Data set]. Retrieved June 7, 2026, from https://ec.europa.eu/eurostat/databrowser/view/hlth_ehis_bm1e

Eurostat. (n.d.-b). Obesity rate by body mass index (BMI) [Data set]. Retrieved June 7, 2026, from https://ec.europa.eu/eurostat/databrowser/view/sdg_02_10